

Cloud Computing Tutorial - 2

Demonstration of a PaaS Application: Creating a Trailblazer App on Salesforce Trailhead

Observation:

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Tutorial-2

Demonstration of PaaS Application:
creating a Trailblazer app on Salesforce Trailhead.

① What is PaaS?

soln: Definition: PaaS is a cloud service model that delivers a complete development and deployment environment in the cloud, including infrastructure, middleware, databases and development tools.

- you manage your application and data.

Benefits:

- ① Faster development: No need to configure servers, built-in development tools, quick deployment
- ② Cost Effective: Pay-as-you-go pricing, No hardware purchase, reduced maintenance costs
- ③ Scalability: Automatic scaling based on traffic, handles high load easily
- ④ Simplified development: One-click deployment, CI/CD integrations available.
- ⑤ Better collaboration - integrated version control system.
- ⑥ Security & updates managed - reduced operational burden.

Eg: Salesforce, AWS elastic beanstalk, Google app engine, ...

Trailhead: official online learning platform by salesforce technologies through interactive modules, hands-on exercises, and real projects. Helps developers, administrators and users learn salesforce skills.

Steps to build apps using Salesforce lightning:

① Use lightning app builder

- drag and drop interface with features like creating pages, adding components, designing dashboard

② Create lightning components:

Developers build reusable UI elements called lightning components. Such as buttons, forms,

③ Connect to Salesforce data

2. Setting up a Trailblazer playground.

① Trailblazer playground is a personal Salesforce developer environment provided through Salesforce Trailhead where users can practice, test and build applications safely.

- Sign up for Salesforce Trailhead and launch the playground

3. Creating trailblazer app

- Building basic app in Salesforce

- creating custom objects

Eg: Student object with fields like email, name, course.

- adding fields and data types

Eg: datatype for course is text and email is email

- designing UI through No-code or low-code development methods.

4. Automating processes in trailblazer app

What is process automation in Salesforce?

- Process automation in Salesforce is the use of built-in tools to automatically execute business processes based on specific triggers, conditions or events.

Eg: When a new student record is created, send welcome email automatically.

This saves time and manual effort, reduces human error and improves workflow efficiency.

Flow builder: used to create complex workflows using a visual interface. Automates record updates, approvals...

Eg: creating a flow for automatically sending welcome email when new student record is created.

- defining trigger, add action, configure email, save and activate

Process builder: another tool to automatically perform actions when certain conditions are met. Now, Flowbuilder is encouraged compared to process builder.

5. Deploying & Testing the Trailblazer app.

Developers test the app to verify that automation rules, flows and components behave as expected.

Steps to test:

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① Test the Features

② - Forms accept correct data, automation rules or run properly, records are created or updated correctly

② use debug logs;

to check for errors, flow execution, etc

③ debug flow builder

④ use developer console

sandbox: It is a copy of salesforce production environment used for testing, development and debugging without affecting real data.

Q: How companies use salesforce PaaS for business apps?

Ans: allows companies to build, customize, and deploy business applications on the cloud without managing infrastructure.

CRM, Sales automation, customer support systems, business process automation, customer enterprise applications, integration with other systems.

Salesforce Trailhead Demonstration:

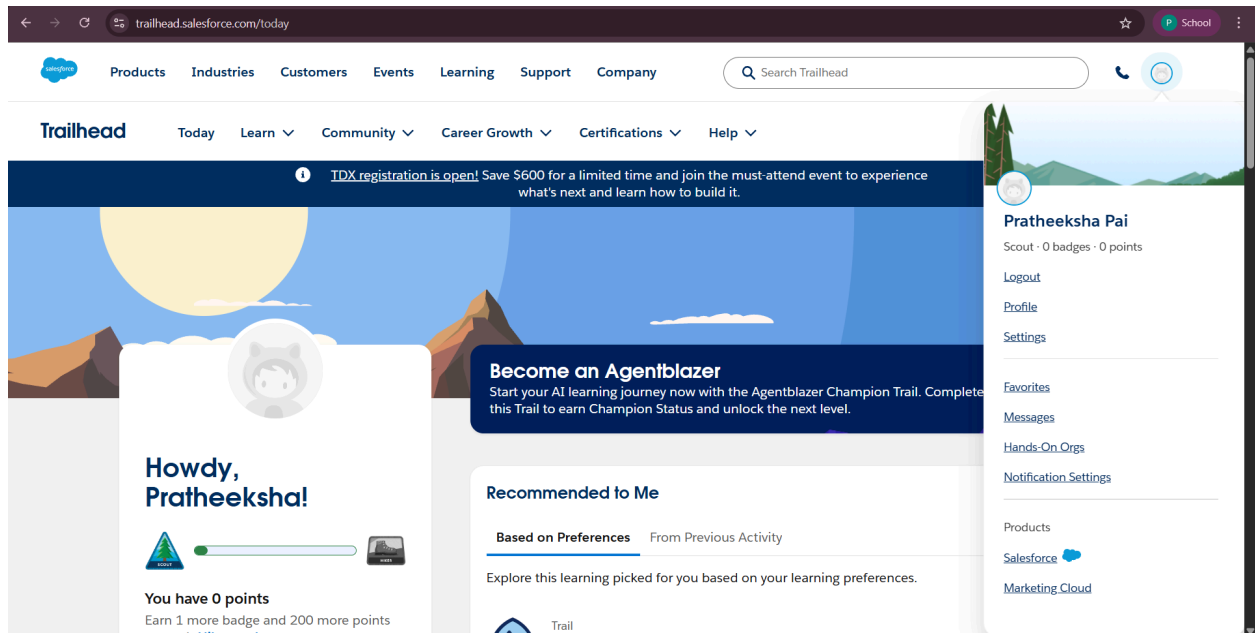


Fig 1.1: Salesforce Trailhead HomePage

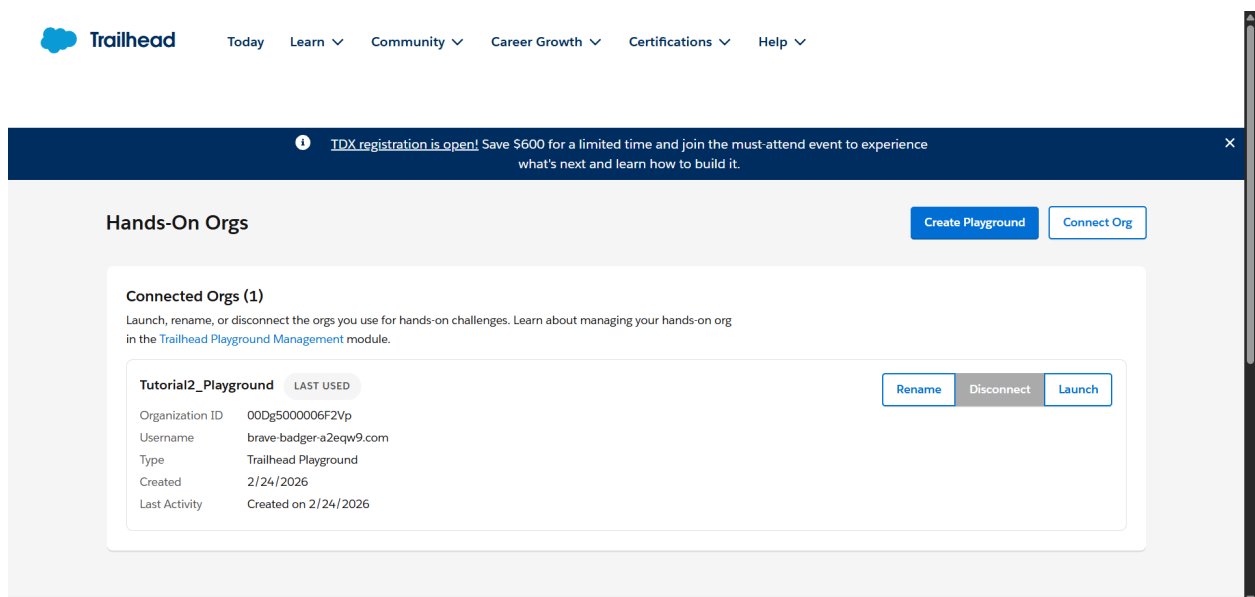


Fig 1.2: Playground creation

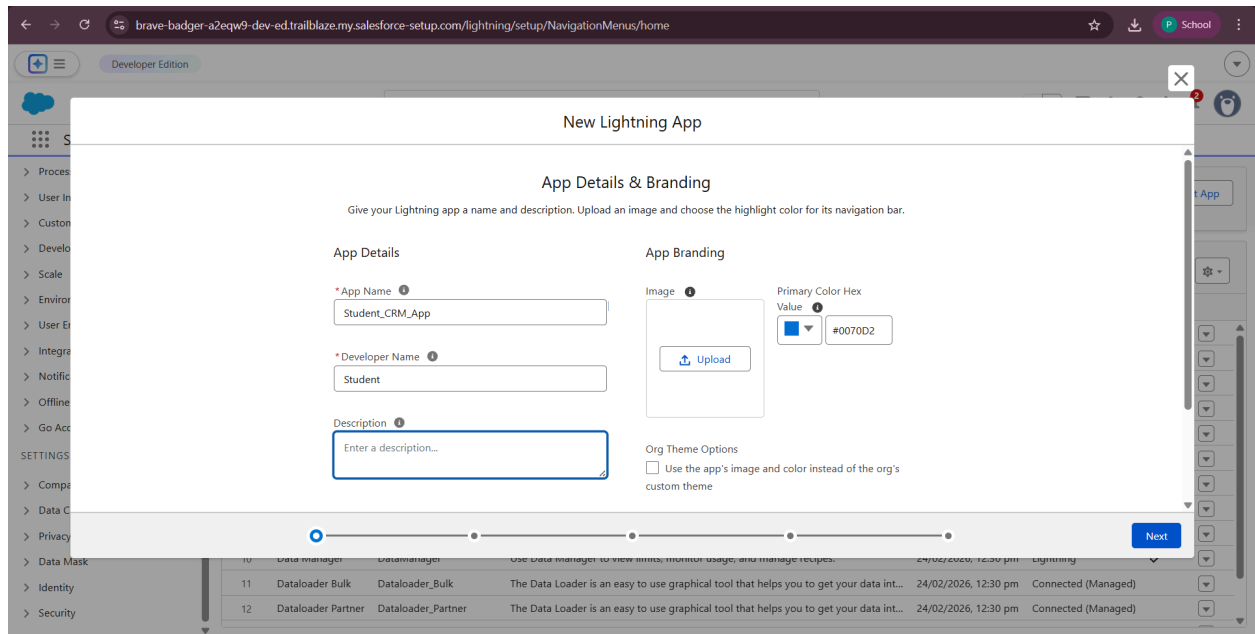


Fig 1.3: New Lightning App creation

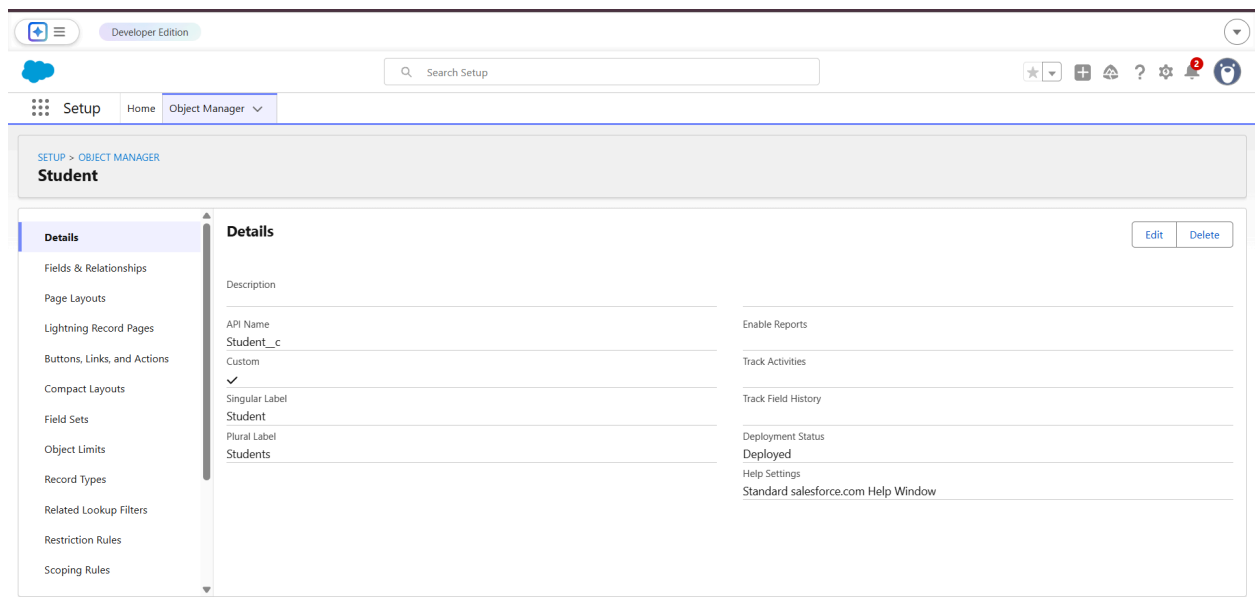


Fig 1.4: Adding a custom object - Student Object creation

SETUP > OBJECT MANAGER
Student

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules

Student
New Custom Field

Step 2. Enter the details Step 2 of 4

Field Label

Length

Field Name

Description

Help Text

Required ☐ Always require a value in this field in order to save a record

Unique ☐ Do not allow duplicate values

☐ Treat "ABC" and "abc" as duplicate values (case insensitive)

☐ Treat "ABC" and "abc" as different values (case sensitive)

Fig 1.5: Adding a custom field

SETUP > OBJECT MANAGER
Student

Details
Fields & Relationships
Page Layouts
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Scoping Rules

Fields & Relationships
6 Items, Sorted by Field Label

Quick Find

[New](#) [Deleted Fields](#) [Field Dependencies](#) [Set History Tracking](#)

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Course	Course_c	Text(50)		☐
Created By	CreatedById	Lookup(User)		☐
Email	Email_c	Email		☐
Last Modified By	LastModifiedById	Lookup(User)		☐
Owner	OwnerId	Lookup(User,Group)		☒
Student Name	Name	Text(80)		☒

Fig 1.6: All fields and relationships

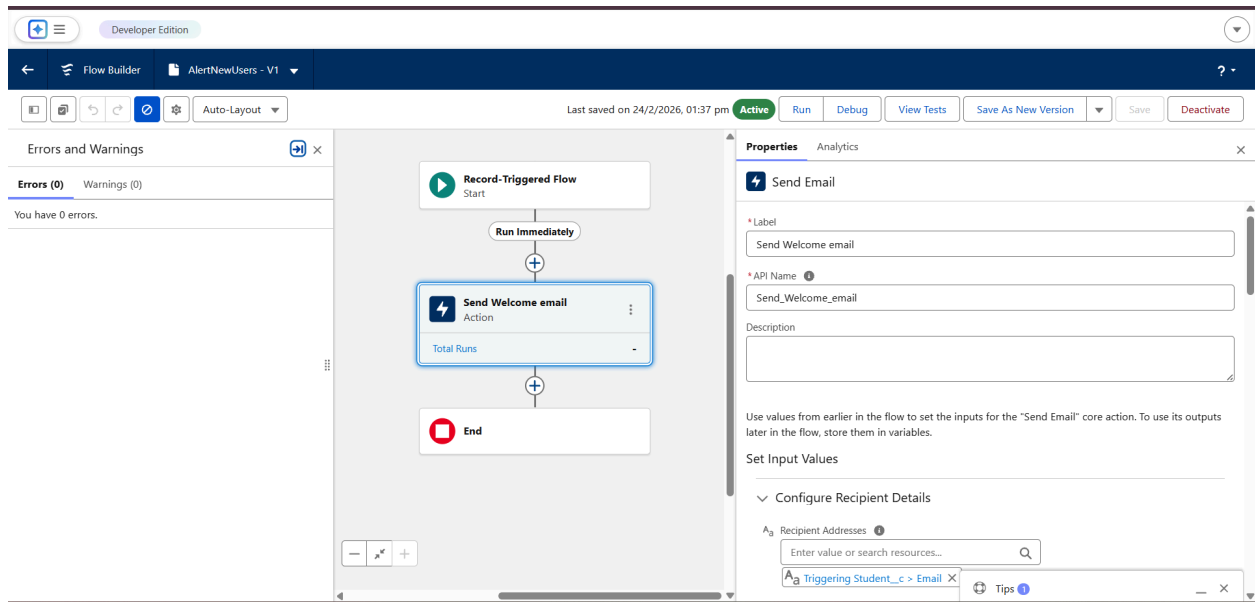


Fig 1.5: Flow Builder flow creation - setting up automation rules

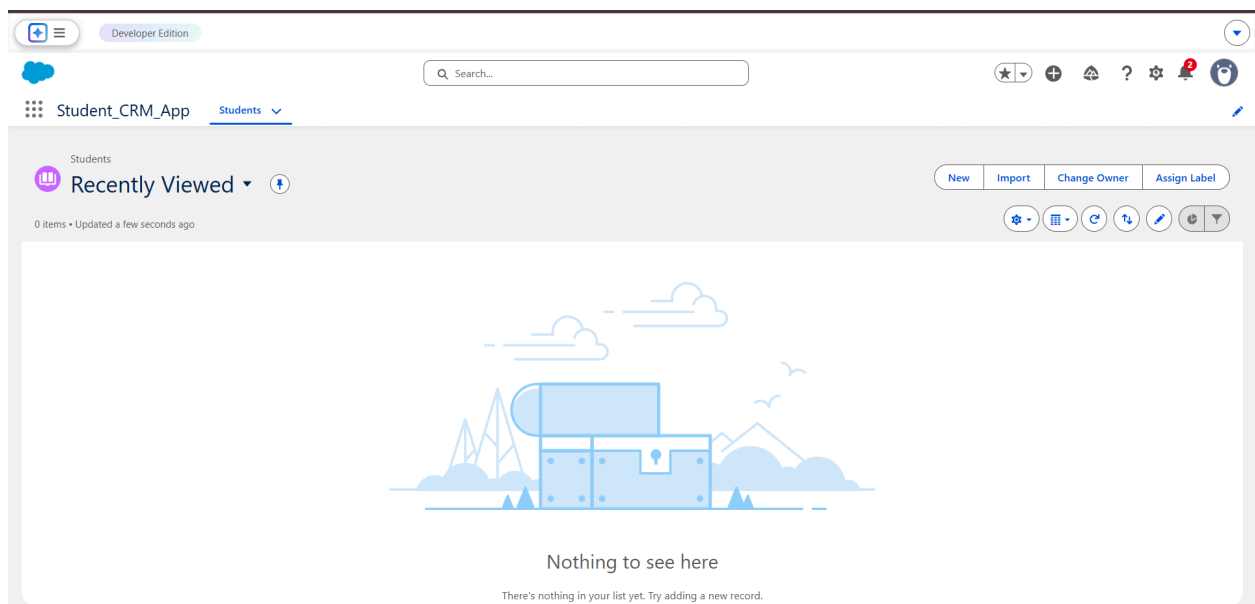


Fig 1.6: Newly created Sudent_CRM_App

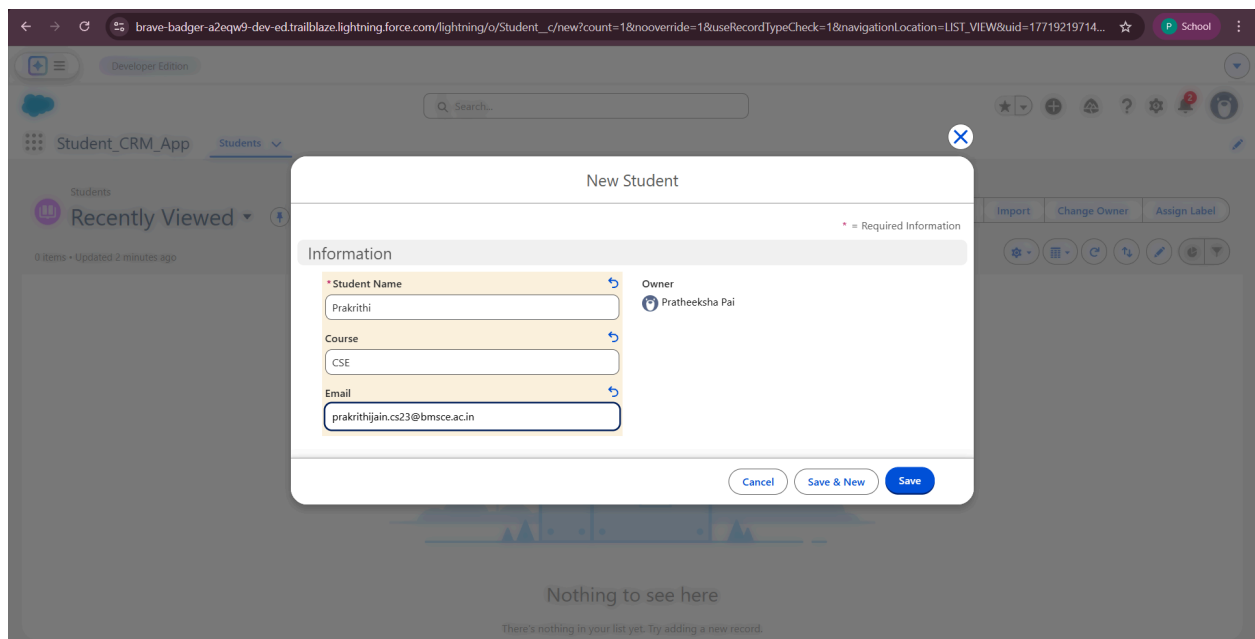


Fig 1.7: Creating New Student

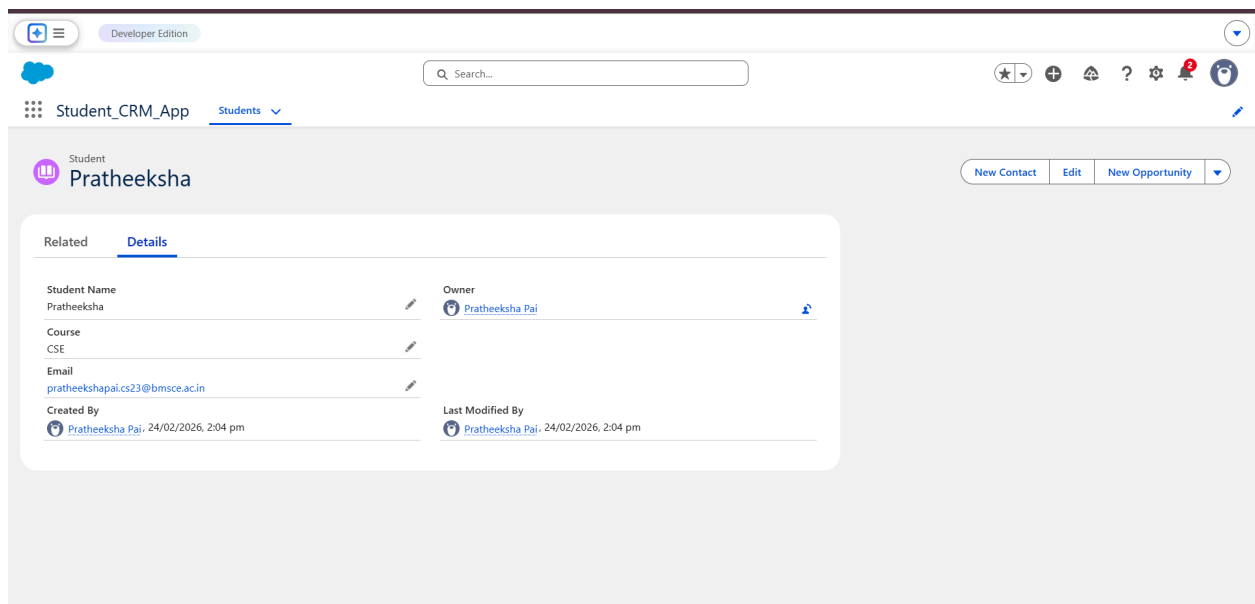


Fig 1.8: New Student Record created

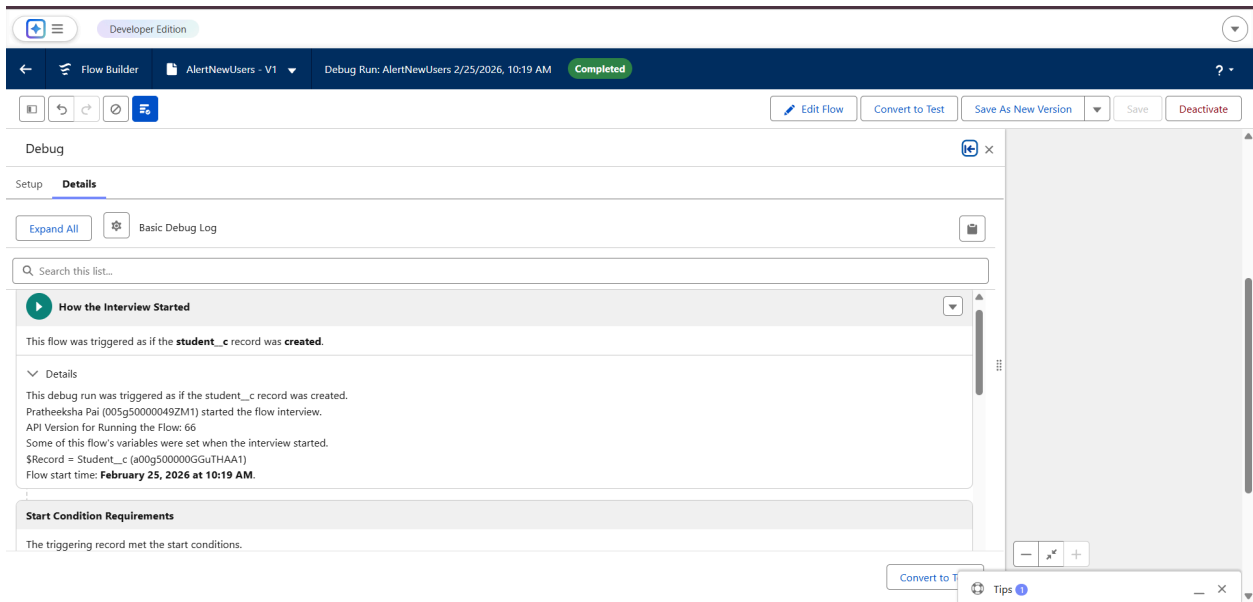


Fig 1.9: Testing and Debugging the app

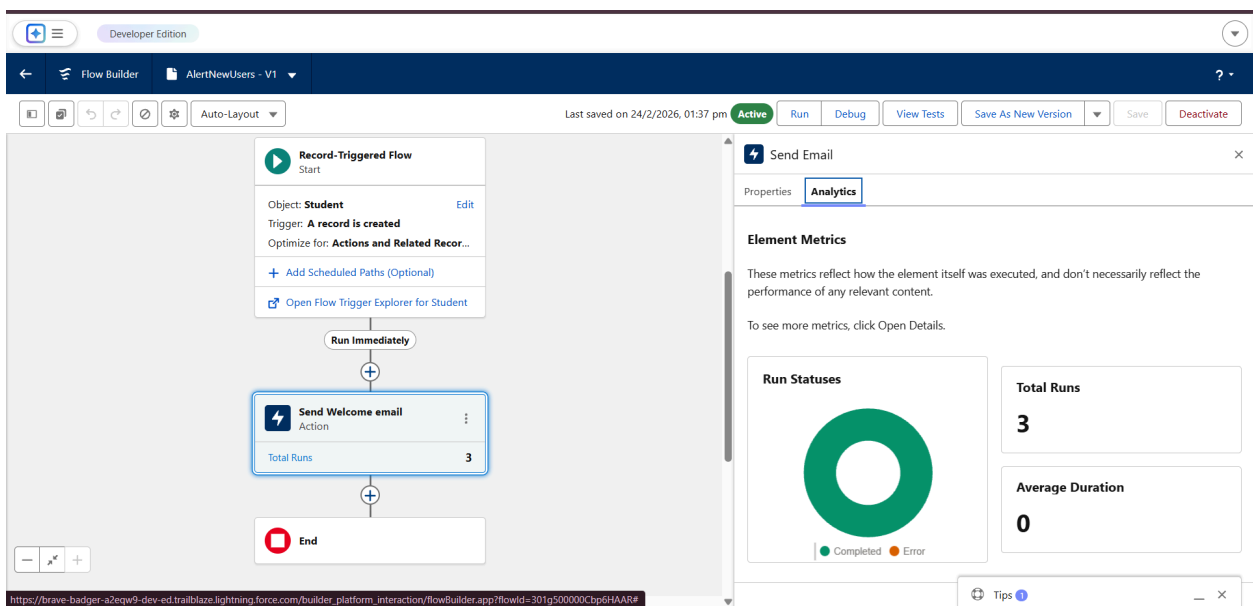


Fig 1.10: Student_CRM_App analytics